simon-5501-08-slides

2025-10-14

Topics to be covered

- What you will learn
  - The two-sample t-test
  - The t-distribution
  - Critical values and p-values
  - R code for the t-test
  - Confidence intervals
  - Sample size justification
  - R code for sample size justification
  - Your homework

Population model

- Population 1
  - $X_{11}, X_{12}, \ldots, X_{1N_1}$
  - X_{1i} are independent $N(\mu_1, \sigma_1)$
- Population 2
  - $X_{21}, X_{22}, \ldots, X_{2N_2}$
  - X_{2i} are independent $N(\mu_2, \sigma_2)$

Speaker notes

The two sample t-test is based on a population model where there are N_1 observations in the first population and N_2 observations in the second population. In general, the size of the two populations, N_1 and N_2 are assumed to be very large.

Sample values

- Sample 1
  - $X_{11}, X_{12}, \ldots, X_{1n_1}$
  - Calculate $\bar{X}_1$ and S_1
- Sample 2
  - $X_{21}, X_{22}, \ldots, X_{2n_2}$
  - Calculate $\bar{X}_2$ and S_2

Speaker notes

Because the populations are so large, you need to take a sample (hopefully a representative sample) from each population. With the sample, you can calculate sample statistics.

Hypothesis and test statistic

- $H_0 : \ \mu_1 - \mu_2 = 0$
- $H_1 : \ \mu_1 - \mu_2 \neq 0$
  - Accept H_0 if $\bar{X}_1 - \bar{X}_2$ is close to zero

Speaker notes

The null hypothesis for the two-sample t-test is that the population means, μ_1 and μ_2 are equal, which is the same as saying that the difference between the two population means is equal to zero.

The population means are unknown, but you can use the sample means,

How close is close?

- $T = \frac{\bar{X}_1 - \bar{X}_2}{se}$
- se = standard error
  - $se = S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$
- S_p = Pooled standard deviation - $S_p = \sqrt{\frac{n_1\ S_1^2 + n_2\ S_2^2}{n_1 + n_2}}$
- Not valid with heterogeneity
  - That is, $\sigma_1 \neq \sigma_2$

Speaker notes

You measure how close $\bar{X}_1 - \bar{X}_2$ is to zero by using the standard error. The standard error is a measure of how much sampling error you have when using $\bar{X}_1 - \bar{X}_2$ to estimate $\mu_1 - \mu_2$.

This standard error relies on equal variation in both groups. You'll hear more discussion of this issue later in the presentation.

Break #1

- What you have learned
  - The two-sample t-test
- What’s coming next
  - The t-distribution

The t distribution

- $T = \frac{\bar{X}_1 - \bar{X}_2}{se}$
  - Variation in the numerator AND the denominator
  - Use a t-distribution, not a normal distribution
    - $df = n_1 + n_2 - 2$

Speaker notes

The test statistic, T, is the ratio of the difference in sample means to the standard error. This statistic has variation both in the numerator and the denominator. This produces a statistic that is not normally distributed, but close to normal. It is the t-distribution.

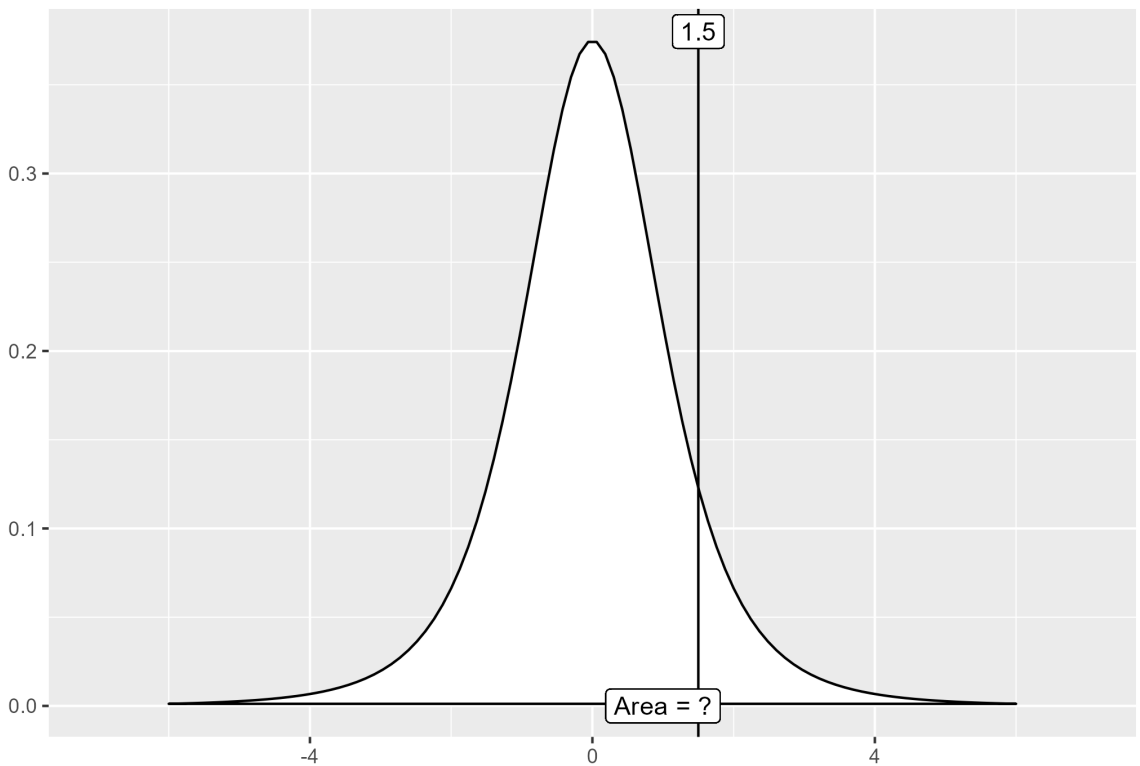
The t-distribution has degrees of freedom. A large degrees of freedom means very little sampling error in the denominator. It is the total sample size ($n_1 + n_2$) minus two degrees of freedom associated with the two estimated means used in the standard deviation calculation.

Comparing the t and normal distributions

Speaker notes

This graph compares the normal distribution to a t distribution with 10 degrees of freedom and a t distribution with 4 degrees of freedom. Both the normal and the t-distributions are symmetric. The t-distributions have a little bit less probability near zero and a bit more probability at the extremes.

P[t(4) < 1.5]



Simon, 2024-10-06

```
1 pt(1.5, 4)
```

```
[1] 0.896
```

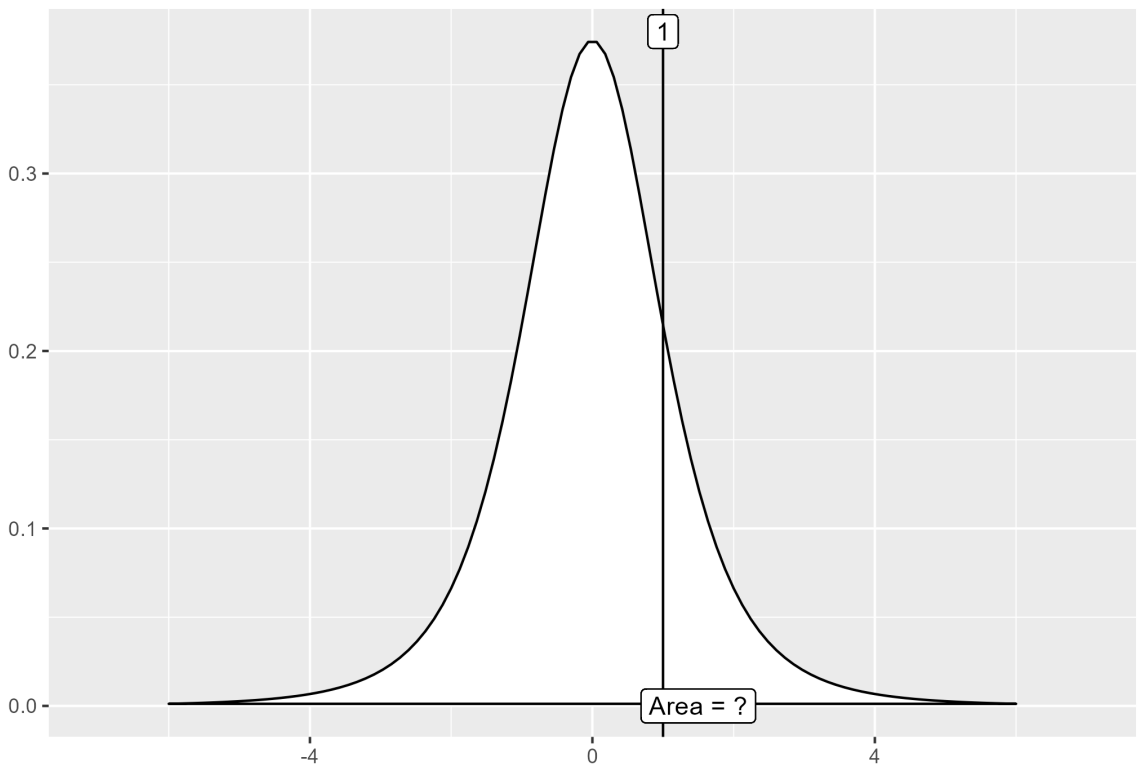
```
1 #
2 # Compare this
3 # to the normal
4 # probability
5 #
6 pnorm(1.5)
```

```
[1] 0.9331928
```

Speaker notes

The calculations of normal probabilities used the pnorm function. For the t-distribution, you should use the pt function. Specify the limit, in this case 1.5, and the degrees of freedom. The probability that a t-distribution with 4 degrees of freedom is less than 1.5 is a bit less than 90%. This is slightly different from the probability you would get with the normal distribution.

P[t(4) > 1]



Simon, 2024-10-06

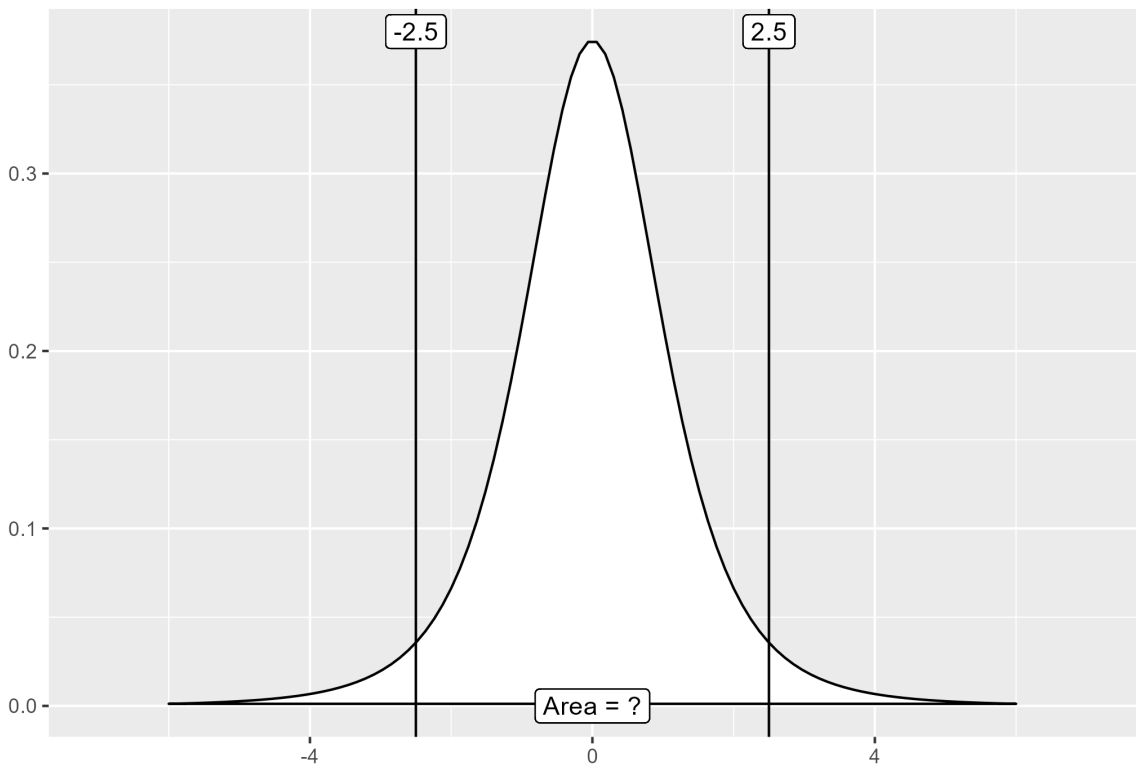
```
1  1-pt(1, 4)
```

```
[1] 0.1869505
```

Speaker notes

To get the probability or area to the right, subtract the pt function from 1. The probability that a t-distribution with 4 degrees of freedom is greater than 1 is about 19%.

P[-2.5 < t(4) < 2.5]



Simon, 2024-10-06

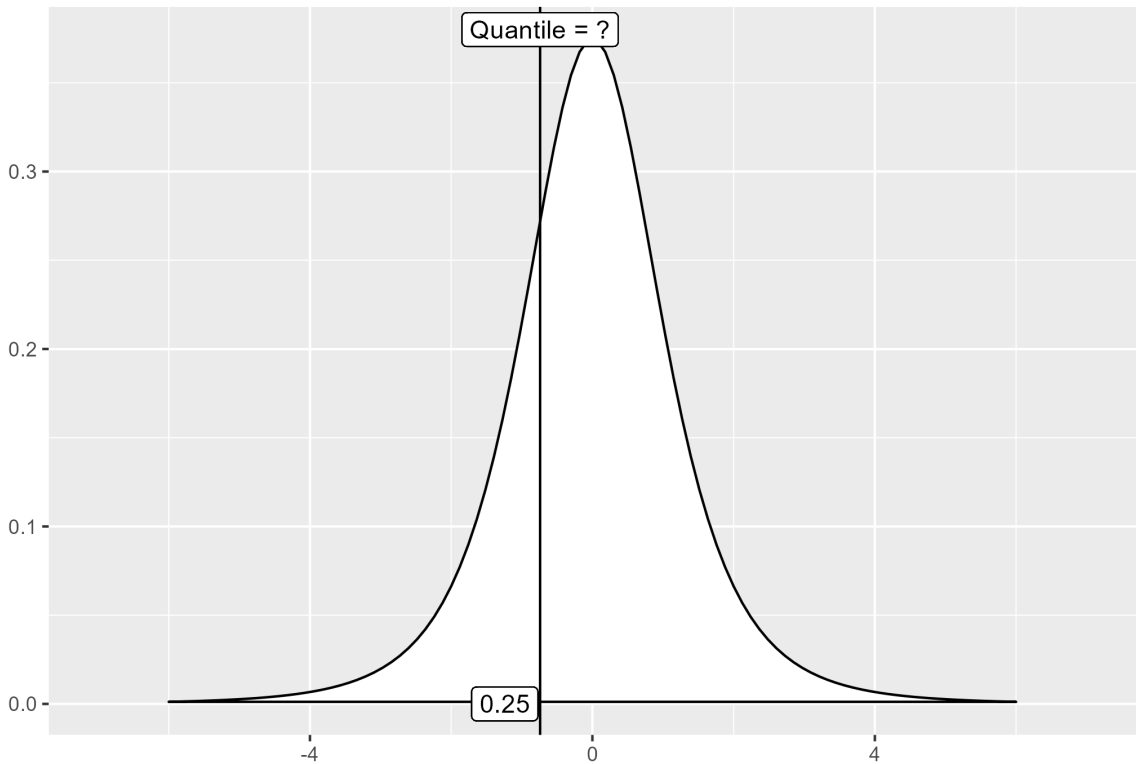
```
1  pt(2.5, 4)-pt(-2.5, 4)
```

```
[1] 0.9332335
```

Speaker notes

To get the probability of being between two numbers, compute the difference between pt functions. The probability that a t-distribution with 4 degrees of freedom is between plus or minus 2.5 is about 93%.

25th percentile of t(4)



Simon, 2024-10-06

```
1 qt(0.25, 4)
```

```
[1] -0.7406971
```

Speaker notes

Use the quantile function, qt, to calculate any quantile of the t-distribution. The lower quartle (25th percentile) of a t-distribution with 4 degrees of freedom is -0.74. About one-fourth of the probability is less than this value and about three=fourths of the probability is greater than this value.

Break #2

- What you have learned
  - The t-distribution
- What’s coming next
  - Critical values and p-values

Type I and Type II errors

- Type I error, Rejecting the null hypothesis when the null hypothesis is true
  - α is the probability of a Type I error
- Type II error, Accepting the null hypothesis when the null hypothesis is false
  - β is the probability of a Type II error
  - Power = 1 - β

Speaker notes

Let’s review the definition of Type I and Type II error rates.

Critical values, 1

- $H_0: \ \mu_1 - \mu_2 = 0$
- $H_1: \ \mu_1 - \mu_2 \neq 0$
  - $T = \frac{\bar{X}_1 - \bar{X}_2}{se}$
  - Accept H_0 if $t(\alpha/2, df) < T < t(1 - \alpha/2, df)$

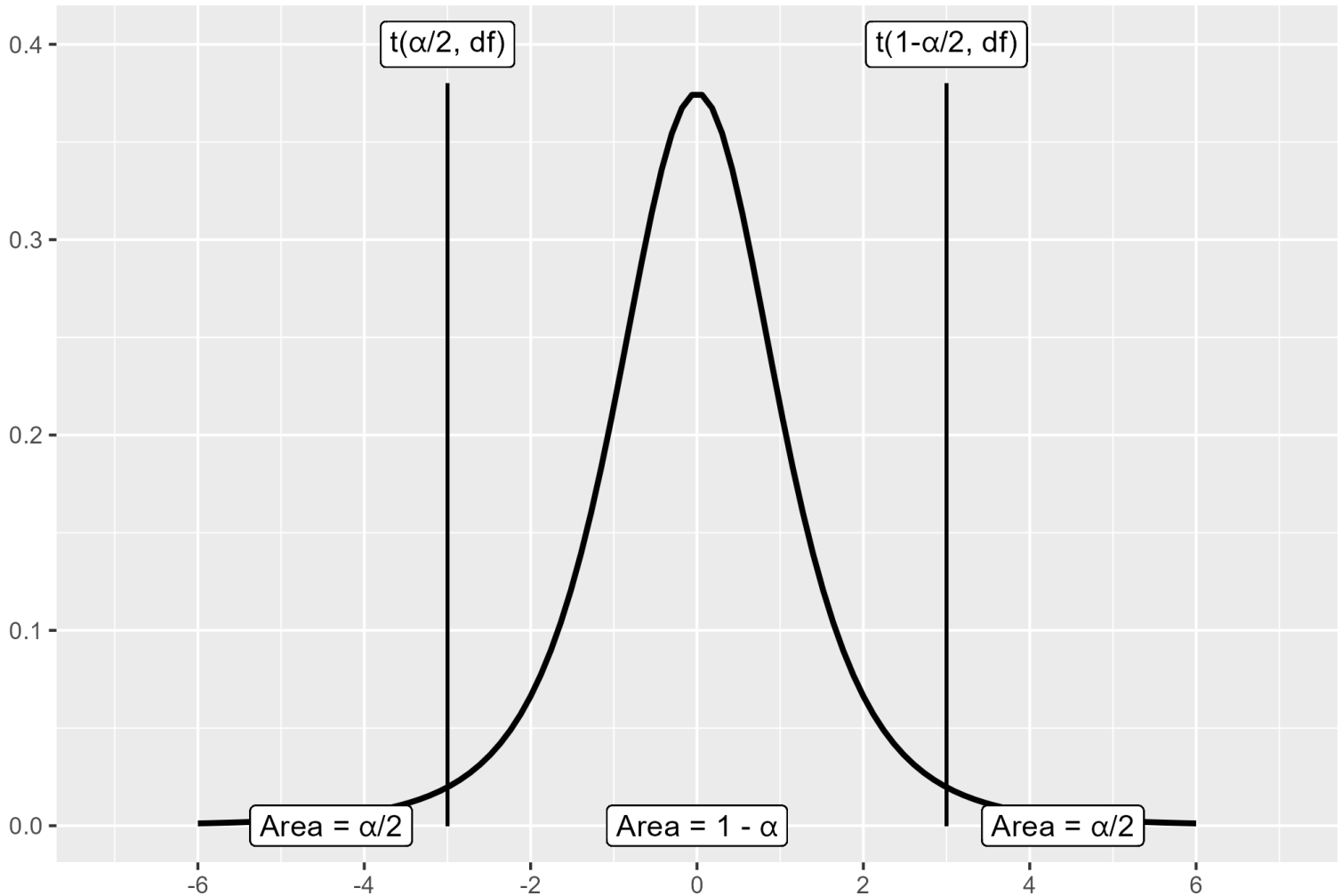
Speaker notes

The formal test of hypothesis looks at whether the test statistic T is close to zero. Close means that it falls between the $\alpha/2$ and $1-\alpha/2$ percentiles of a t distribution with n1 + n2 -2 degrees of freedom.

The standard error is, in general terms, an estimated amount of variation for statistics computed from a sample. In this case, the standard error is an estimate of how much variation to expect when you compute the difference between two sample means.

You use the t-distribution because there is randomness in the test statistic in both the numerator and the denominator. The numerator uses means calculated from the sample. I'm not showing the formula for the standard error here, but it uses standard deviations calculated from the sample.

Critical values, 2



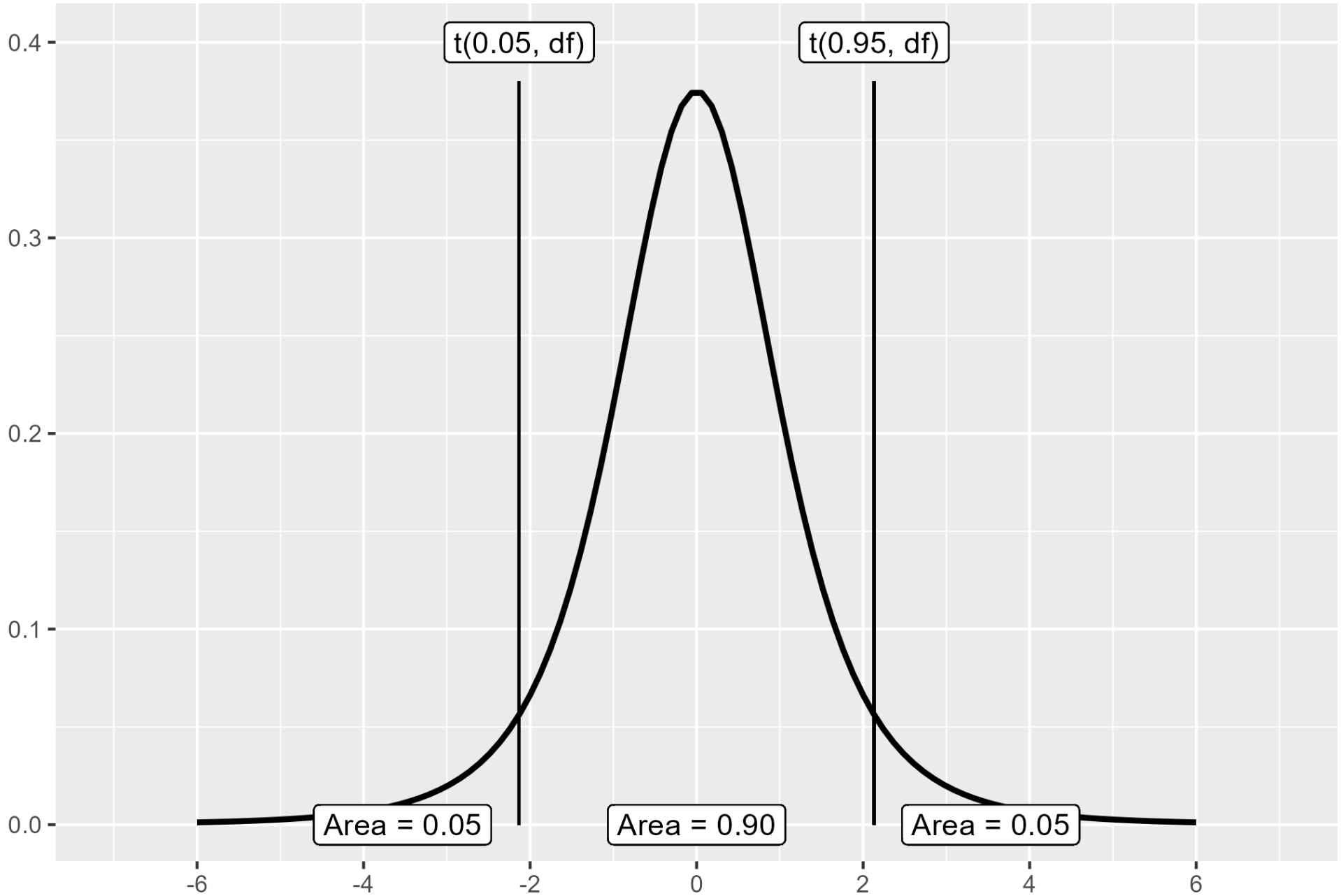
Simon, 2024-10-06

Speaker notes

Here is a graphical display of the critical values. Close to zero is equivalent to having a probability of 1-alpha in the middle of the t-distribution. A probability of 1-alpha in the middle and alpha/2 plus alpha/2 in the extremes is equivalent to a Type I error rate of alpha.

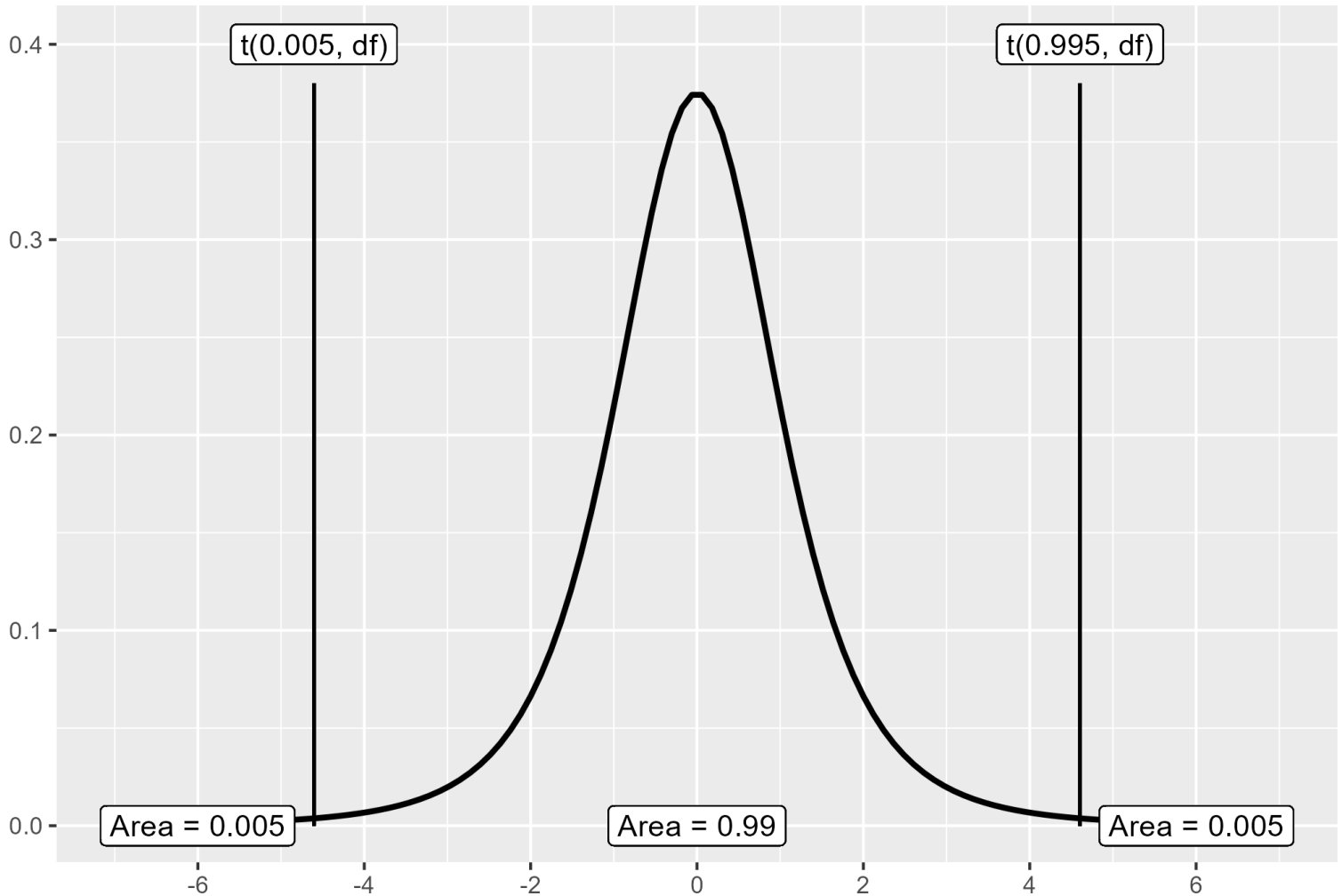
You want alpha to be small. What are some possible values for alpha?

Critical values, 3



Simon, 2024-10-06

Critical values, 4

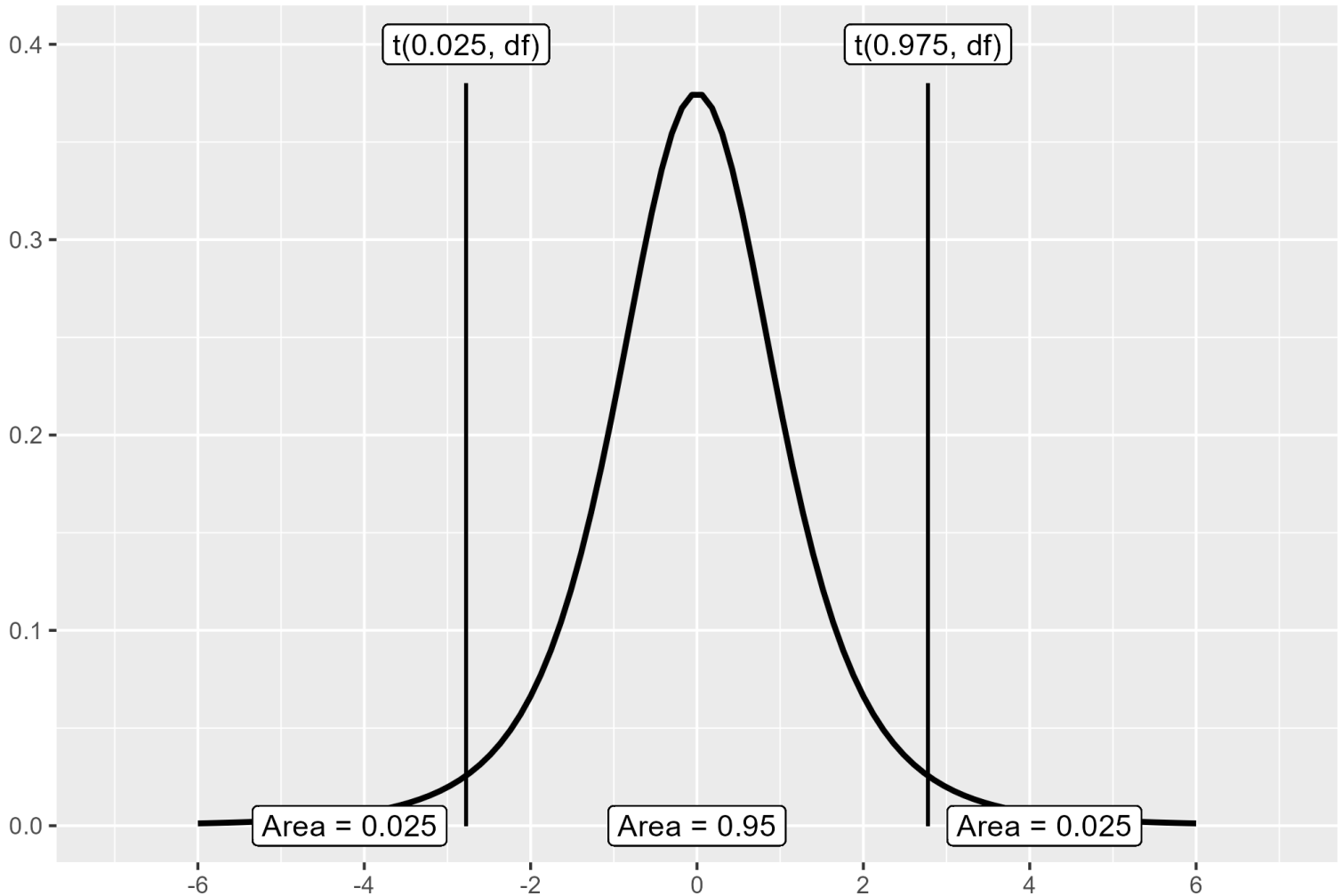


Simon, 2024-10-06

Speaker notes

Here is a set of critical values that places 99% of the probability in the middle and 1% at the extremes. These critical values would produce a Type I error rate of 1%.

Critical values, 5



Simon, 2024-10-06

Speaker notes

Here is a set of critical values that places 95% of the probability in the middle, 2.5% on the low end, and 2.5% on the high end. These critical values would produce a Type I error rate of 5%.

Which critical values should you choose?

- If a Type I error is far more serious than a Type II error,
  - Use t(0.005, df) and t(0.995)
  - P[Type I error] = 0.01 (conservative)
- If a Type II error is far more serious than a Type I error,
  - Use t(0.05, df) and t(0.95)
  - P[Type I error] = 0.10 (liberal)
- If you want to avoid controversy
  - Use t(0.025, df) and t(0.975, df)
  - P[Type I error] = 0.05
  - Lemming school of research

Speaker notes

Which critical values should you choose? This is an important question, and one that is often ignored.

There are settings where a Type I error rate is very serious and a Type II error rate is less serious. Perhaps there is a large financial burden associated with rejecting the null hypothesis, and a false positive finding would lead to a lot of wasted money. In such a case, having a small Type I error rate, 0.01, may be attractive. For these settings, use critical values of t(0.005, df) and t(0.995, df).

You will often see the word “conservative” applied to tests that have a very small Type I error rate. This is not at all related to politics, though.

The reverse may be true. Perhaps rejecting the null hypothesis leads to a small financial burden, but accepting the null hypothesis is where you see a large financial burden. In such a case, you may tolerate a larger Type I error rate, 0.10. For these settings, use critical values of t(0.05, df) and t(0.95, df).

In contrast to the conservative Type I error rate, a larger value for the Type I error rate like 10% is sometimes described as “liberal”. Again, there is no political implications in the use of this word.

To be honest, most researchers do not think too hard about the financial implications of a Type I error rate. They just go along with what everyone else does and set the probability of a Type I error rate to 0.05. This leads to critical values of t(0.025, df) and t(0.975).

I call this the Lemming school of research. At one time or another, a parent will try to discourage their children from bowing to peer pressure with the warning “If all your friends jumped off a cliff, would you jump off also?”

P-value

- p-value = $2P[t(n_1 + n_2 - 2) > |T|]$
  - Probability of sample results or results more extreme
  - Accept H_0 if p-value > α
- Why the 2?
  - Measuring extremity in either direction

Speaker notes

The p-value is a simple way to test hypotheses that removes the need to calculate critical values. The p-value is quite ubiquitous. It is by far the most commonly computed statistics in research. It is also quite controversial. Some researchers have called for an outright ban on p-values. More have called for the research community to rely less on p-values and more on other statistics like the confidence interval.

Don’t worry so much about the controversies now.

There are two big advantages to the p-value. First, it is easy to apply. A small p-value (less than alpha) means that you should reject the null hypothesis. A large p-value (greater than alpha) means that you should accept the null hypothesis.

Second, when you report the p-value, someone with a statistically conservative outlook can see if the p-value is less than 0.01. Someone with a statistically liberal outlook can see if the p-value is less than 0.10.

Postural sway data, 1

```
# A tibble: 2 × 4
  Age     fb_mn fb_sd     n
  <chr>   <dbl> <dbl> <int>
1 Elderly  26.3  9.77     9
2 Young    18.1  4.09     8
```

Speaker notes

Here are some descriptive statistics from a small study looking at postural sway. Patients stood on a force plate that could measure how much the patient swayed front-to-back and side-to-side. A large amount of sway could mean problems with loss of balance and falls. The sample size was small, nine elderly patients and eight young patients. There does seem to be more front to back sway in the elderly patients with an average of 26 compared to 18. But the sample size is small and there is a fair amount of variation in the patients, with standard deviations of 9.8 and 4.1.

Let’s show some of the calculations that go into testing the hypothesis that the population mean front-to-back sway is the same for both groups.

Postural sway data, 2

- Calculate pooled standard deviation
  - $S_p = \sqrt{\frac{n_1\ S_1^2 + n_2\ S_2^2}{n_1 + n_2}}$
  - $S_p = \sqrt{\frac{9\ (9.77)^2 + 8\ (4.09)^2}{9 + 8}}$
  - $S_p = 7.64$

Speaker notes

The first step is to compute a pooled standard deviation. The pooled standard deviation is a composite of the two individual standard deviations. It will always lie somewhere between the two individual standard deviations. In this example, you need to square the two standard deviations, multiply each by their respective sample sizes, divide by the total sample size and then square root the whole thing.

Phew! A lot of work and we are just getting started.

Postural sway data, 3

- Calculate standard error
  - se = $S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$
  - se = $7.64 \sqrt{\frac{1}{9} + \frac{1}{8}}$
  - se = 3.71

Speaker notes

Next you need to compute the standard error. The standard error is, in general terms, a measure of variation for a sample statistic. In this case, it is a measure of variation for the difference between the two means. This formula is also a bit messy, but notice that standard error is in some sense inversely related to the two sample sizes. The larger the sample sizes, the smaller the standard error. This should make sense. Larger sample sizes lead to greater precision.

Postural sway data, 4

- Calculate T
  - T = $\frac{\bar{X}_1 - \bar{X}_2}{se}$
  - T = $\frac{26.3 - 18.1}{3.72}$
  - T = 2.2

Speaker notes

Now compute the test statistic, which is the difference in the sample means divided by the standard error. If this test statistic is close to zero, that implies that the difference that you are seeing between the two sample means can easily be explained by sampling error. When the test statistic is large negative or large positive, the difference between sampling means cannot not be explained away by sampling error.

The value that you get, 2.2, is borderline. It is not too close to zero, but it is not too far away either.

Postural sway data, 5

- Calculate degrees of freedom
  - df = n1 + n2 - 2
- Calculate critical values
  - t(0.025, 15) = qt(0.025, 15) = -2.13
  - t(0.975, 15) = qt(0.975, 15) = 2.13
  - Since T is outside the two critical values, reject H_0

Speaker notes

You calculate the degrees of freedom by adding up the two sample sizes, but you have to subtract two. You lose two degrees of freedom because you are estimating variation around two different sample means. In this example, because the sample sizes are small, the degrees of freedom is ony 15.

The critical values are the values that delineate what it means to be close to zero. The 0.025 quantile and the 0.975 quantile of the t-distribution with 15 degrees of freedom are plus or minus 2.13.

The test statistic, 2.2 is outside of the two critical values, making the test statistic far away from zero, but just barely. This means that you should reject the null hypothesis and conclude that the front-to-back postural sway in the population of elderly patients is larger than the postural sway in the population of young patients.

Postural sway data, 6

- Calculate p-value
  - p.value = $P[t(n_1 + n_2 - 2) > |T|]$
  - p.value = $2\ P[t(15) > |2.2|]$ = 2*(1-pt(2.2, 15)) = 0.044

Speaker notes

You could also test the hypothesis about the two population means using a p-value. In this case, the p-value is twice the probability that a t-distribution with 15 degrees of freedom is larger than the absolute value of the test statistic. You use the pt function to calculate probabilities for the t-distribution. It is one minus because you are looking at a greater than probability and you double it because you are interested in both very large positive and very large negative test statistics.

Postural sway data, 7

```
    Two Sample t-test

data:  FBSway by Age
t = 2.2044, df = 15, p-value = 0.04353
alternative hypothesis: true difference in means between group Elderly and group
Young is not equal to 0
95 percent confidence interval:
  0.2715453 16.1451214
sample estimates:
mean in group Elderly   mean in group Young
             26.33333              18.12500
```

Speaker notes

Well, I don’t know about you, but I am exhausted after all this work. There was nothing terribly complex about this. A few additions, and subtractions, a few fractions, and toss in a couple of square roots. It’s all more tedious than it is difficult. The only hard part is remembering all the formulas.

The good news is that while I suffered doing these calculations, you don’t have to suffer. You can get R to do all the work for you. Here is what R presents when you use the t.test function.

At the bottom are the two sample means. Above that is the 95% confidence interval, which you encounter later in today’s lecture. Near the top is the test statistic, 2.2, the degrees of freedom, 15, and the p-value, 0.044. Notice that I am rounding in my interpretations.

Because the p-value is small, less than 0.05, you would reject the null hypothesis. Let’s interpret this result in the next slide.

A proper interpretation

- $H_0 : \ \mu_1 - \mu_2 = 0$
- $H_1 : \ \mu_1 - \mu_2 \neq 0$
  - Reject the null hypothesis and conclude that the mean front-to-back sway in the population of elderly patients is greater than the mean front-to-back sway in the population of young patients.
- What to look for in the interpretation
  - A statement about the population, not the sample
  - Use of directional words

Speaker notes

Remmeber that the hypothesis is a statement about the two populations, the population of elderly patients and the population of young patients. These two populations are very large, and we only have enough resources to get sample of nine in one group and eight in the other group. Even so, the difference that we saw in the sample means was larger, larger than could be accounted for by sampling error.

When you make a statement, don’t say that the sample mean for one group is equal to or different from the sample mean for the other group. Your hypothesis uses Greek letters and concerns the population parameters. Notice that we use words like “mean front-to-back sway in the population of elderly patients”. Also use directional words. You did reject the null hypothesis, so you know that there is statistical evidence that the two population means are not equal. But go a bit further and explain the direction of the difference. Notice the word “larger” in the interpretation. You could have also said “greater than”. if you flip things around you could talk about how the mean sway is smaller in the population of young patients.

If you accepted the null hypothesis, you would use words like “equal” of “the same as”.

Break #3

- What you have learned
  - Critical values and p-values
- What’s coming next
  - R code for the t-test

Live demo

This will be a live demonstration of part 1 of simon-5501-08-demo. This program uses the dataset postural-sway.tsv. Refer to the data dictionary for information about this dataset.

Break #4

- What you have learned
  - R code for the t-test
- What’s coming next
  - Confidence intervals

Confidence interval for difference in means

- $\bar{X}_1 - \bar{X}_2 \pm t(1 - \alpha/2, n_1 + n_2 - 2)se$
  - Range of plausible values for $\mu_1 - \mu_2$

Speaker notes

The formula for a confidence interval for the difference between two means is shown here. The calculations are tedious, but not difficult.

Interpretation

- Statement about population mean difference ($\mu_1 - \mu_2$)
- Range of plausible values
- Not a probability statement
  - 95% confidence does not mean 95% probability
- If you collected 100 independent samples,
  - Roughly 95 would contain $\mu_1 - \mu_2$

Using a confidence interval to test a hypothesis

- A confidence interval is a statement about $\mu_1 - \mu_2$
  - Accept H_0 if interval contains zero
  - Reject $H + 0$ if interval does not contain zero

Example with the postural sway data

```
    Two Sample t-test

data:  FBSway by Age
t = 2.2044, df = 15, p-value = 0.04353
alternative hypothesis: true difference in means between group Elderly and group
Young is not equal to 0
95 percent confidence interval:
  0.2715453 16.1451214
sample estimates:
mean in group Elderly   mean in group Young
             26.33333              18.12500
```

We are 95% confident that the difference in front-to-back sway between the population means for elderly and young patients is somewhere between 0.3 and 16.1. This is a wide interval, which is not unexpected for such a small sample size. The

Break #5

- What you have learned
  - Confidence intervals
- What’s coming next
  - Sample size justification

Bad joke about sample size

- How many research subjects does it take to screw in a light bulb?

Speaker notes

Here’s a bad joke about sample size. How many research subjects does it take to screw in a light bulb? At least 300 if you want the bulb to have adequate power.

Another bad joke

- A researcher gets a six-year, ten million dollar grant and writes up a report after the grant is over.

Speaker notes

Here’s another bad joke. A researcher gets a six-year, ten million dollar grant and writes up a report after the grant is over. “This is a new and innovative surgical procedure and we are 95% confident that the cure rate is somewhere between 3% and 98%.

Sometimes I will tell a joke during a consulting session and it sails over the head of my client. The two statistics in a bar is a joke that sometimes requires a bit of experience with Statistics before it makes sense. But this joke always lands. It always makes people laugh. This is why sample size calculations are so important.

Two ways to justify your sample size

- For studies without a research hypothesis
  - Show that your confidence intervals will be reasonably narrow
  - Not too narrow
- For studies that have a research hypothesis
  - Show that your Type I and Type II error rates are both small
  - Equivalently, show that the power is large

Speaker notes

If you came to me with a question about sample size, the first thing I ask is “What is your research hypothesis”. Most times, you already has a hypothesis in mind. Sometimes, though, you and I have to brainstorm together to come up with a research hypothesis.

Other times, the work that you have in mind does not lend itself well to a research hypothesis. Studies of diagnostic tests, for example, normally do not have a formal research hypothesis. In these settings, you can often justify the sample size by showing that any confidence intervals that you produce will be reasonably narrow.

If you do have a research hypothesis, then you can justify your sample size by showing that it will produce a study where both the Type I and Type II error rates are small. This is often stated not with a small Type II error rate, but a large power. Power is defined as one minuss the Type II error rate.

Three things you need to justify your sample size

1. Research hypothesis
2. Measure of variability
3. Minimum clinically important difference (MCID)

Speaker notes

If you came into my office asking about a sample size, I would start with asking about your research hypothesis. That’s the easy one.

Let’s presume that your hypothesis involves the comparison of outcomes for two groups. They might be drug versus placebo, exposed versus unexposed. It might be a comparison of two demographic groups: young versus old patients, males versus females.

Next I would ask you about a measure of variability for your outcome measure. If it is a continuous outcome, then I would need you to provide me with a measure of variability for your outcome measure. A standard deviation is best, but other measures of variation, like the range, would also be okay.

Finally, you need to define the minimum clinically important difference. How much of a difference is big enough to convince you that the two groups differ in a way that is important.

Scenario

- Replicate postural sway study
  - Different populations
  - Same outcome measure
- Research hypothesis, $H_0\ \mu_1 - \mu_2 = 0$
- Standard deviations: 9.77, 4.09
- MCID = 4

Scenario, R code

```r
sample_size_estimate <- power.t.test(
  n=NULL,
  delta=4,
  sd=9.8,
  sig.level=0.05,
  power=0.9,
  type="two.sample",
  alternative="two.sided")
```

Scenario, Output

```
     Two-sample t test power calculation

              n = 127.1097
          delta = 4
             sd = 9.8
      sig.level = 0.05
          power = 0.9
    alternative = two.sided

NOTE: n is number in *each* group
```

Break #6

- What you have learned
  - Sample size justification
- What’s coming next
  - R code for sample size justification

Live demo

This will be a live demonstration of part 2 of simon-5501-08-demo.

Break #7

- What you have learned
  - R code for sample size justification
- What's coming next
  - Your homework

simon-5501-08-directions.md

Refer to the programming assignment on my github site.

Summary

- What you have learned
  - The two-sample t-test
  - The t-distribution
  - Critical values and p-values
  - R code for the t-test
  - Confidence intervals
  - Sample size justification
  - R code for sample size justification
  - Your homework